ActInf Livestream #021.2 ~ John Boik

Livestream | John Boik | 2:02:17

hello everyone
welcome thanks for joining actin flab
this
is live stream number 21.2
i think and this is our sixth discussion
with john the first four were background
discussions where
john and i worked through these jam
boards
and through the three-part series that
john had written
science-driven societal transformation
and
now here we are today on may 11th
2021 in our second participatory group
discussion so this is going to be
an unpacking and an extension and an
opportunity to
share perspectives and ask john
questions
so thanks to all who have joined live
and who will be joining live
and also if you're watching on youtube
feel free to leave a question in the
chat and we will
relay that to you so we're just going to
go around
with a quick round of introductions
maybe each of us could just say hello
and something that they're thinking of
or something that they're excited to
talk about
today and then we will pass back to
john and just start off on this road
together so

i'm daniel and i'm in california i will
pass
to lee
hi uh i'm lee i'm in i'm in london at
the minute
um although i'm a phd researcher at the
university of york
uh john i'm today i'm interested to talk
about
um maybe a minimum viable prototype for
one of the clubs um and also uh
a little bit around the the metaphor of
the super organism a bit as well if we
have jan
uh and our path to uh scott scott i
think
thank you my name is scott david i'm at
the university of washington
applied physics lab where i'm the
director
of information risk and synthetic
intelligence
research initiative delighted to be here
today
and i'd also i'm very very interested in
chatting about
the scale scale independent variables
and
how the interventions at one scale
can help to alter
system performances at other scales due
to fractal nature of
some of the affordances we're talking
about great interest to me
and i'll pass it to dave douglas i'm
in the philippines 120 miles north of
manila where it's cool because it's a
mile high city

and i just came across the answer to a
question what is a state
in the parcels and particles paper dr
fristen
says a state is a particle
so yes it really really is monads not
only all the way up but all the way down
to and uh who's left daniel you've
already we already know you
we'll go to we'll go to john thanks for
the the spicy introduction dave
i'm john boyk um
if you go to set two
slide one uh this
this series this these numerous talks
are on the topic of
science-driven societal transformation
that's the title of a
three-part series that i recently
published in the journal sustainability
um part one is world view part two
is motivation and strategy and part
three and three is design
i i i'm looking at societal
transformation through a particular lens
in this series and that lens is the de
novo
development of
fundamentally new systems so when i say
systems i'm talking
about the economic systems governance
systems legal systems
et cetera the
we're going to focus today i think on
maybe some of the practical
aspects like how would this be done
and uh i would like to just start by
saying just a few words about

this the kind of theory of change that i
talk about especially in
paper number two if i were
brilliant which i'm not and i came up
with this amazing new design for
societal systems new governance systems
etc
if there were no way to actually
practically implement that
then i it would have just been an
academic exercise
so part of this series is arguing that
there actually is
a reasonable way to move forward a a
viable path to move forward that is
doable that
that actually could be put into into
implementation
so this the series is is itself is
is essentially um outlining an r d
program for the development of this
framework to study and uh create and
develop and test
new integrated systems and that happens
at according to this theory of change
that happens at the
at the local level so the concept is is
that this r d
program is a is a collaboration between
the
global science community and
local communities and the new systems
are implemented at the local level on a
volunteer basis in what i call a club
model
so um the the concept there is that the
scientific community would have
developed

the tools and technologies and resources
and
things necessary to make these new
systems work well
uh and um they the
local communities some volunteer local
communities somewhere in the world would
would look at the results of all the
computer simulations and all the studies
that have been done and say that looks
interesting
gee it looks like our community might
really benefit from this
um uh let's give it a field trial let's
give it a try
and uh so that's the partnership and
then the
community itself now this could be i
don't mean a city i don't mean a
nation i really mean like perhaps a
subset of a city so
say just a thousand people or something
like that it might be enough
to start a club and the club would
function everyone in the club would
function
as normal in society except that on top
of
on top of you know there's another layer
of organization on top of the normal
layer of organization
and that's this club level so civic club
you can you can think of it as a
civic one and the civic club
has all these you know uh
ways that they go about decision making
and exchanging
currencies and various things that allow

them
then to test out these new frameworks of
of uh societal systems so that's the
gist of it and i just want to say a few
things about this club model before we
really get going just so we've kind of
all on the same page on that um
just before i do that maybe go to slide
two of that same deck
i'll point people to uh principled
societies project dot org if
they're looking for more information on
this
all of our videos are posted there and
the papers are posted there and there's
some other materials posted there
so for for information go to psp
and let's go to slide 17.
okay
oh that is not the right uh not
quite the right one i wanted the theory
of change one and that's
not it
[Music]
uh which slide is that you're looking at
seventeen one yep
there you change you know that's somehow
i'm uh oh
oh here we go
yes there is change and strategy uh
uh this is just kind of uh
capturing what i just said the idea is
to implement new
fundamentally new systems at the club
level
uh i'll call the club a community but
this community could be
relatively small than just a portion of

the city and the reason i'm
one of the reasons i'm looking at the
club level is because i see it as kind
of the goldilocks level it's the
it's the smallest size
organization of a community that could
make the club
able to replicate right so and and
function
so if the if we were instead just
looking at say an individual business
that business like i say say i was
promoting a cooperative business model
or something like that
that business would have to function
within the wider
set of businesses and would have to
compete in the
normal market for the consumer's
attention
and uh it would have a hard time doing
that because it's operating
according to some different set of rules
in a normal business
so the at the club level i see that as
just
large enough that all of these new
functional systems can be tested and
and it's large enough that the the
organization the community
can be self-replicating the club model
can be self-replicating
it can grow in a local area and the idea
can spread to other local areas
um uh all right
uh slide 19 please can we just go to
just lee and then scott just because
we're getting hands raised on each of

these slides so yeah so lee and then
scott
sure
john what are the um in order to start a
club
what are the minimum requirements that
are needed do they need any funding or
are there other set of conditions that
it might need
um ideally there wouldn't be any
requirements ideally there would be
support
from the from the you know the global
community to start a club but
but but the but the clubs are designed
such or at least
later we'll today we'll talk about a
simulation i did of a of a club model
a particular framework i called the lida
framework
but i designed that so that it could
bootstrap its own growth
so there was no it was not necessary
to you know for it for uh to reason
to have millions of dollars say to start
the club or anything like that it really
just grew
step by step exponentially fast so in
the first few years hardly anything was
happening
and then it started to grow you know
very rapidly after that
uh so you know what would be necessary
mostly just the
individuals that are that look at the
the
by the time a a field trial would ever
be started

there would have been you know let's say
hundreds of
papers scientific papers and simulations
and all other
aspects of this so everyone would know
what they're getting into this would be
very clear what the
what the you know what the new systems
are and what they do and how they
function
and then at that point all that would
really be necessary is for some
community somewhere in the world
to say oh that looks really interesting
let's do a field trial you know here's
we have a thousand people in our little
in our community that are interested in
this and let's give it a shot
so that's the main thing i mean maybe a
little bit of funding would be necessary
but not an extreme amount
so it sounds oh it sounds almost like
there's a prelude
phase which is the what we're in and
then
the stage will be set for proper
emergence for a community to then take
that next step
but um it's a really interesting
question about the viable
system so i hope that we pursue that
line as well
yeah john anything else and i did i just
maybe we'll add that by the time the
first field trial ever starts
there really should be no surprises
about any of this i mean
the community would you know conduct the

trial
but it should be a foregone conclusion
what's going to happen
as far as you know how it works and what
the benefits are
because this should have been worked out
very well already so
uh you know it shouldn't be as it's not
like uh here's a
you know here here's a new
here's a new machine a tractor or
something
now see if you can drive it it's not
quite like that because there would be
uh you know training involved we would
realize what the purpose of the tractor
is what
we're going to accomplish with the
tractor how it works how you repair it
you know like it would be a uh
again there's just should be very few
surprises by the time the first field
trial starts
that doesn't mean a community wouldn't
be able to make decisions on how it
what it funds and you know how it
functions of course the community
would make decisions about how about its
the kinds of jobs it creates and things
like that
but the fundamental mechanics mechanics
of the system should be pretty well
worked out
just very quickly then so is the um is
the particular community so you've got
this prelude phase where you're kind of
working out all the details of the de
novo design

is that done with the community oh
partly i mean that would be with any
communities
like globally in a sense that prelude
part is the global science community and
any
individuals or communities anywhere in
the world that want to participate
provide feedback do user testing
um you know all the little parts of the
systems would be also tested in various
groups and
there would be like really just a ton of
testing that would occur before the
before the first formal field trial ever
started
um and the and that particular community
what could have input on
um you know what kind of particular
design elements it might want in its
system
so and and that's actually a good point
that i'm glad you said that because
i also wanted to say that the purpose of
this r d
program is not to build one system that
fits
all it's to build it's it's an ongoing
exploration of
them you know different pieces and
approaches and
ways you might create systems
such that they fulfill their purpose so
there could be a
wide variety of you know
parts to choose from or ways to do
things or
you know hopefully there would be

numerous kinds and types
and pieces of systems that would be
tested over time that communities could
you know implement then and all with the
purpose of
of uh you know fit for purpose
you know all with it with the concept of
fit fit for purpose behind us
got and then anyone else who raises
their hands
i just wanted to go back you were
talking a little bit about the size of
groups
in the clubs it's a the optimization's
going to be really interesting because
you might have
and dunbar's number involved there that
dumbar's number concept being
the most people you can have as your
friend
kind of idea so whatever is 120 or
something but
one of the things that's interesting
also you have the internal voice
dynamics of the club plus the external
voice dynamics of the club
so as the club coheres an internal voice
it will have a more coherent external
voice which will allow it to entrain
and harmonically couple other systems
more effectively right because it'll be
more clear here and
one of the things that i'd like to get
into later in the conversation is we
have uh
some ideas we've been working on on
scaling how
legislative judicial and executive

function the three functions of all
governance systems have rule making
enforcement of the rules and operation
of the rules how those
naturally layer on just as a function of
scale
as you increase in scale from rule
making you have to go into different
abstractions for
enforcement and operation under those
rules and so that
would be very interesting to look at the
dynamics of club growth
as an internal process of internal voice
and a recognition of the layering on of
governance that occurs
internally and then how that governance
then
needs to expand into the external
environment
in order to be more effective the the
bottom line
is if you have a system a club where we
say
we are the free speech club we have the
rights to free speech
and there's no other club that has a
duty to respect that then you don't have
a right to free speech outside of the
club
right and say how do you iterate the
story
outside of the club so that it becomes a
reality that's shared so it can be
enjoyed
right right i'd like to talk about that
yeah yeah there's there's a there's a
ton there that we

go through uh i have these concepts in
mind
as i was designing these systems or
designing the frameworks so
all good points um just a little bit on
size i'll try to
talk about a couple of things briefly
that you raised a little bit on size
um so so in
our brains we have a community of
neurons that are
that are functioning to help the body
make decisions and
that that community of neurons is very
much connected with the rest of the body
also
feet and the liver and the heart and
everything else
well how does all of that you know how
does
how do all those cells and tissues and
organisms
cooperate together to make the body
function the way the body functions
one thing that happens is there's a very
rich stream of communication that is
passing between
all these different cells and tissues
and things for hormonal signals and
small molecule signals and blood is
circulating and there's all these
this awash in signals and communication
so uh communication
is one of the prerequisites for
any kind of large organized
you know instead of individuals
whether they're cells or whether they're
people to function together

so one of the questions that we can talk
about today
um is what kinds of tools are necessary
to allow for rich communication to occur
in say a group of let's just say 10 000
people or a hundred thousand people or a
million people
how do how does a large group of people
come together
richly communicate their narrative their
story
their their both you know
i want to tell i want to tell my group
what is happening for me
what i anticipate will happen if we
choose a
over b over c uh where this might lead
in the short run where this might lead
in the
long run how this fits into other issues
issues d and e and f i mean you know i'm
an intelligent person
everyone else has intelligence and um
that intelligence can be used in the
group
uh cognitive you know the group
cognition
in order for the group to make wise
choices about
you know actions and and where they want
to put their attention
so so the tool question is a big one
and i would say that that currently
there are no tools
that allow a large group of individuals
to come together to collaborate with
via rich communication and rich
storytelling

right we that doesn't exist yet i would
say and i have ideas on how we might
create that but that's one of the
that's one of the one of the tasks that
would have to happen
over the over the first several years of
the program
is addressing that particular problem of
how does a group richly communicate
and collaborate and making decisions so
so if we're just talking to get to your
point now
if we're if we just have a small group
and we're just
just talking we're working out things
we're making decisions together
well maybe that group can't be more than
a few people before it starts to fall
apart or you know maybe it can't be more
than 10 or 20 or 100 people or something
um
but if we have some way to richly
communicate then maybe we can scale that
you know we can scale that communication
uh to a larger group to thousands or ten
thousands
and still there might be some limit be
beyond what a club wouldn't
you know like maybe a club should never
be more than a hundred thousand or
never more than a million or never more
than ten thousand you know who knows
and so that's between that's of
individuals in a club
and then there should be similar
communication between clubs
that must happen and that's getting to a
that's touching another point that you

raised is
how do how does a club reach reach the
you know interact with the wider world
and
the the both and and the
my framing both with other clubs how
how does decision making occur between
clubs how does communication happen
between clubs and then also how does
communication happen with
the rest of the world thanks john
um stephen and then scott and then
anyone else who raises their hand
yeah thanks i i think i'm interested by
this um
the word purpose you know we think about
purpose and all of this
and clubs tend to have a particular
uh focus or purpose as you're mentioning
and
i think it's interesting you're bringing
in the science
intentionality you know the
intentionality of the science community
to enable
sort of seeding so i'm wondering how
that might you might think about that
sitting alongside say
community organizations or like
community hubs for instance that or
multi-service agencies
in the community or nodes of support
those types of things this might be
something which could fit in there and
give this kind of
club purpose that might
complement or is it different
similar to some of those types of

service provisions
that exist out in towns and cities
i'm not exactly sure
the question you're pointing to but let
me say a few things and maybe i'm going
to touch on it
um so so
um kind of in the abstract
the the you know the concept behind this
whole
program is that um
that that an organism has an intrinsic
purpose and that trans
intrinsic purpose is to thrive both now
and
in the future so expected thriving and
uh in order to do that it uses cognition
right and via cognition and we've talked
about how active inference fits in
to that whole picture via cognition it
really reduces the uncertainty about its
capacity to thrive it's
the essential variables that it you know
that it that it needs that it
functions the world it functions in and
whether it's need true needs will be met
right so
uh that's an organism and i'm viewing a
society as an organism right
so in the sort of the larger picture the
abstract
little abstract the
question is you know is is a community
functioning in a way is a new system
functioning in a way that is reducing
uncertainty about the capacity to thrive
so that's that's uh that would be
yeah that would be you know uh

implemented in a way via some kind of
fitness metrics
so just like in normal society in a
regular society we have say gdp
and other various types of you know
information we gather and metrics that
we use there would be a set of metrics
that would go along with this whole
program so that a community would know
how it's doing and how it will
how it can expect to be doing two years
from now and five years from now and
ten years from now you know maybe even
20 years from now
so so there may be many
subgroups within say that say that
there's a club and the club is 10 000
people
for example clearly there would be some
groups within that club that would be
interested in various
aspects of you know
topics or ways of doing things or
various kinds of communication
and you know maybe some some
subgroups are really focused on the
social
components of the community components
the you know maybe emotional components
social components
and then other groups are interested in
technical components and
modeling or whatever you know other
other groups are interested in the
environmental components for example
well that's great that's you know
association between groups is a part of
cognition our neurons associate

in in ways that allows for cognition of
the whole
and in the same way a you know club in a
community would do the same
so there could be all kinds of subgroups
that are all functioning but the
umbrella purpose would be present right
because it would be reported daily say
or
even hourly or weekly you know there
would be everyone would know how
how we're doing as a whole and that
we're all in this together
to raise our our joint
capacity to thrive and and maintain our
maintain our ability to thrive so yes
there would be all kinds of groups that
would focus on different
aspects of this and the umbrella
you know reporting and collecting of
information would be happening still
across groups across the community so
that there would be some sense of how
are we doing
and what's going to happen next you can
i just reply
slightly to that yeah just very good
yeah just that's really interesting and
it sounds as well
this could tie in with the kind of
another type of evaluative capacity
or ongoing rolling evaluative capacity
but with an agent based or
active inference based
engine so to speak so that it's giving
this
update to communities around
how they are progressing

so to speak when that may help them in
other areas some of which may be
outside of the club's
more bounded area you know it gets into
the niche and complexity
so the interesting if you thought about
this might be
how this might link in with some of the
new developments in evaluative
evaluation going on and that might be
quite a good avenue
yeah there would be i mean if you were
going to collect information
uh and and assess how we're doing and
what's going to happen next there would
be all kinds of
evaluation that would have to occur not
just on the simple things like you could
think of some simple things right off
the bat
like do we have enough food do we have
enough water well fine but humans are
complex
organisms we have complex needs social
needs
you know emotional needs you know like
physical needs
you know we and all of those needs
should be taken into account
in in some kind of you know
comprehensive evaluation of how are we
doing and what's going to happen next
and and obviously every any organism has
to know that you know
every organism wants to know it needs to
assess its status like how
am i how am i doing am i too cold am i
too hot am i hap

unhappy am i happy you know like every
organism needs to do that
and a community a community as an
organism needs to do that
cool thanks john also welcome blue and
on the the goldilocks
note and this narrative note i really
like what scott had said about the
internal and
external communications and as internal
communications
become structured from a connections
perspective but also from a narrative
perspective
the external interfaces and narratives
can be structured so goldilocks
doesn't just evoke the um sort of
upside-down u-curve i mean this is
something that we know from
pharmacology from temperature everything
there's too much and there's too little
and there's a failure mode on either
side just like a bowling alley and of
course the middle solution is
going to be better it just wears the
middle but goldilocks also reminds us
that
that middle solution has to be more like
a child's
story rather than some sort of ab truce
optimization
problem so it's kind of a fun reference
that it should be comprehensible
internally and also externally in order
to be functional
right right right so yeah and it may be
just to
touch on that you know from this to say

the scientific viewpoint
one might think about fitness and you
know in terms of
metrics and statistics and things like
that but
other people won't relate maybe so much
to that
kind of assessment so everyone who's
involved has to
there has to be a way to communicate the
ideas of how are we doing and
and you know what is our status
in ways that are meaningful to everyone
involved or else if it's not meaningful
will really won't
work right so so some some people might
might discuss uh you know status in very
technical terms and other people might
discuss status
and more say emotional terms or social
terms or something like that
but but the general sense the general
the general purpose has to make sense to
everyone or else
why be a part of this one other thing on
the goldilocks
is the question is um
to be clear it's not my intent that a
club say a club started in in houston
texas
with a thousand people it's not my
intent
that the club would stay at a thousand
people it would grow
if the whole concept of this is that if
this
this new model this new club model these
new frameworks

actually work if they actually empower a
community to
do better to to have be more secure to
have
a higher quality of life to have
economic security to
you know in numerous ways if these clubs
benefit the population
the way i think it would then they would
grow because it's
kind of like the you know why don't we
use typewriters anymore
because we have computers
that work so much better i mean they're
just so much better
who would want to use a typewriter you
know that kind of thing so which
that's the concept of this it should be
so easy to join a club it should be so
easy to participate
that that like why not
my life will be better if i participate
in a club and if that's
if that comes true if it really works as
advertised
then clubs will grow and the club model
will spread to other places and then
instead of having a club of ten thousand
or one thousand you have a club of a
hundred thousand and
every city has a club and pretty soon
the you know
much of the world is starting to
function along these
you know with these with this along with
these new systems much of the world is
engaging with them or enough of the
world is engaging with them that the

ideas behind them suddenly become the
start to become the dominant ideas
so that the goal is not to help a
community so much the goal is to
actually transform
globally society or at least semi
globally
thanks john i'm gonna ask a question
from the chat and then stephen and lee
so in the chat natalia writes i read in
the paper
thanks for reading the paper natalia
that people would keep their income
when not working i'm curious what role
motivation plays in the success of the
system
and what kinds of structures are in
place to manage that so
agent level motivations and how do we
design incentive structures and
ecosystems for participation
that are able to incentivize various
kinds of participants
all right okay so so what what you know
what motivates us today in our
economy um you know a variety of things
we're complex organisms we're motivated
by a variety of things
but a large one because the way the
system is designed
is that if you don't participate if you
don't go out there and get some kind of
job you're gonna be in the street you
know there's this like
this kind of you know
you're forced to participate and if you
don't and if you don't you're going to
be in trouble

well well that's not very useful uh for
many reasons
far more useful is to offer people what
they actually
really want what what humans tend to
really want which is
a variety of things they want to help
each other they want
to be part of a community they want to
learn they want to develop skills they
want to
do any number of things that engage them
with each other and
in they want to solve problems they want
to be a part of a problem-solving
process they want
to they want to you know cooperate
it makes us happy to cooperate right all
of nature
is cooperating like this cooperation is
the rule rather than the
the exception in nature so how can we
build a system
that just allows us to really be
ourselves to really
get what we want most want right like
i want to be in a in a kind
and cooperative and functional
uh setting a group i want to work with
each other i want to work with the
people around me and
and help each other and they want to
help me and
so how do we make that happen at scale
right that's the question and uh
so the motivation isn't to get rich i
mean we'll talk about the leader
framework in a little while and that's

just one example of a way to organize an
economic system but
but that particular system actually
achieves
uh income equality over time so clearly
the
motivation isn't to make money like
that's not why you're engaged in the
economic system
the economic system is only a it's
really
a uh it's a cognitive system not a not a
money system
money is just a tool that's used as a as
a way to
share information right um so in the
in the leader framework incomes equalize
over time and the motivation to engage
is to be to fulfill all these deeper
human needs that we all have
thanks for the response john so stephen
and then lee
yeah i'll just sort of carrying on a bit
from what you're just saying there
with um the community getting involved
and
their benefits to the purpose and to
their motivation
at the community level um that is a
missing link at the moment
in terms of and it probably was much
more present even 100 years ago
80 years ago you'd have had the women's
institute and you had all sorts of
different
so we professionalized an awful lot of
roles
okay and you see that with nursing and

all sorts of
professions where they've increasingly
professionalized and trained
so the question i've got there
is how do you um
bring in the motivations of the
professionals
who have another niche that they're
trying to optimize around maybe writing
papers getting a career
and things like this and
do they have to be humble
in some way to honor the input from this
community
because obviously you're trying to
invert some of the
assumptions about where everything
should be located i.e
it doesn't need to all be driven by
professions and just keep driving
more professions that you can bring in
community input and wisdom
to also be part of that through these
clubs but how do you
marry that the types of niches that
people are in
and the competition for their time and
right right um i don't i don't actually
see
that as too much of a problem let's just
let's just pick a profession
uh say nursing or something like that
and then the question becomes uh let's
say that i live in houston and the club
has started and
why would i want to you know i'm a
professional i make a
reasonable salary let's say

why would i want to participate in a
club
well there might be numerous reasons i
might see
i might understand the potential of the
club i mean you know there's this whole
body of literature now these simulations
and other literature that
suggests that this club model might
actually empower the
people who belong to a club to make
better decisions
to achieve you know higher environmental
quality
and higher social quality so i might be
interested
uh just to join for that reason because
i see it as a beneficial thing for the
larger community right i can engage with
these people and make decisions and
together we can
you know make beautiful things happen uh
but
let's now let's focus just on the
economic aspect
and again uh you know there what might
be any
number of ways to design any of these
systems economic systems governance
systems you know
there could be all kinds of concepts
that we that the r d program could
assess and try and test out right but of
the in the leader framework the kind of
prototypical economic system that we
hopefully will
get to talking about here um
um the question becomes how do we get

from here to there
like how do we get from today's society
and incomes and structure to
some other structure you know
down the road right so i try to show
that in the simulation that model that i
published
it goes over a 30-year period so
so again the kind of the expansion of
the club and its activities is
exponential
so maybe in the first few years there's
not a whole lot happenings
people join but you know there's not big
effects but then
these it bootstraps itself economically
and otherwise and then they're starting
to be bigger effects
so uh you know just kind of jumping to
the chase a little bit
in the leader framework it's designed so
that incomes rise over time right and
and
the the people in the simulation people
only join
if they're going to be better off
economically otherwise there's no reason
to join in the simulation
so um i i
let's say i'm a nurse i have a pretty
good income why would i join
well if i join at any point
i will get a babes like a um
and my income is high the let's say the
inc
the lowest target income
is rising but my income is still higher
than that

i can still join and i'll get a kind of
a
i'll get a a small a small bonus
uh that's not the word i use in the
paper but it's kind of like that i i'll
get a i'll get a say
i don't know what let's just say a
thousand dollars a year for joining
well that's a thousand dollars a year
that i didn't have otherwise so i
even i get a even though my income is
high relative to where this
income target is growing i still get a
benefit from it and then i get to use
that money and that you know that part
of it is a local
currency so i get to use the dollar
dollar currency the
national currency and this local
currency in the decision making process
and i'm engaged you know
i have i have some skin in the game now
and then as that uh income rises at some
point it's likely to
reach mine or go above what my current
income is and then for sure i want to
join because if i join my incomes you
know like my annual income is going to
rise
maybe substantially so the in the
in the simulation the income target
rises up to about
uh somewhere around 110 the equivalent
of about 110 000
per year per family so
you know that's more than some nurses
make
uh and especially for those nurses that

are at the bottom level
bottom you know economic level or
teachers or
firemen or police or whatever you know
uh there would be motivation to join
because
your income is going to not only
increase but become stable like you
don't have to
if you lose your job there's no worries
you're going to be taken care of this
system
is circulating currency in such a way
that
once you're in you're okay it doesn't
matter if you're working or not
it it's john it's it's a way to rethink
this um
driven by money versus not driven by
money because you've really said it both
ways you've said
people are not motivated by money yet
you've repeatedly appealed to the idea
that every person will gain more money
and implicitly that that's what is
important
and so it's actually just it is what it
is
and that's where active inference and
all these other techniques and extended
cognition come into play
because there'll be a variety of
narratives for people who are in
different
positions so we'll go to lee and then
scott and then i'll ask a question in
the chat
uh yeah so um it's kind of a related

point actually
so uh in the paper um
i remember you were reading that at one
point you're talking about you know
effectively as the dinobo system
uh starts to out compete the native
system you know there are
gonna be winners and losers effectively
so you know if you're a franchise owner
of a
of a mcdonald's or something you know
your cheap labor disappears
so on a larger scale right that means
that
you know they're um
well i guess what i'm what i'm i'm
asking is do you anticipate that
ultimately the current system you know
with
lots of vested interests to remain the
way it is he's going to start pushing
back once it realizes
that this kind of club model is
potentially a threat to it
yeah so right so it's true there's
winners and losers in this
and the and in a sense
if if if you're measuring winners and
losers by
does this civilization survive and
thrive well then there's hardly any
losers because
you know the the whole concept of this
is so that societies and civilization
can thrive into the future right and
that's not
and you know just as a counterpoint
that's not our current system

you know we're our current system is
essentially on a suicide
uh route you know trajectory of
destroying life on the planet so
the current system is not working
uh and you know it's not working in the
sense of our
insecurity about will we even be here in
a hundred years is
is quite high right so uh but but in a
in another sense yes there will be
losers and winners so the idea is that
this r d program is creating and testing
new
ways of organizing society that really
do
that really function well and any
organism that functions well is going to
out compete other organisms you know
that are not functioning well right
in that sense so um a community say that
is using these tools and using these
systems and it's really flourishing
uh probably we'll be doing flourishing
more than other communities that aren't
and that's the reason that other
communities might say hey that looks
good
but let's do that now you have a subset
of say
you know multi-millionaires and
billionaires and
at least on some perspective they will
be losers because
uh you know that that's not gonna
work that's not gonna be compatible with
this new way of doing things
uh and you raised one of the reasons why

i wouldn't be compatible
is that where are they going to get
their low-wage
workers to work and you know work in
their factories and stuff
where are they going to come from if all
the other folks are
having high-wage jobs in these new you
know
these new systems right these new
frameworks so the the business model
will start to falter the conventional
business model will start to
start to falter and as it should because
you know that what what are we what are
we doing here what is the what are we
doing in society right are
don't we want to flourish don't we want
systems that help us to to reduce our
uncertainty and flourish into the future
well i i would think we do overall right
so systems that don't do that really
have to
fall by the wayside or or transform
themselves into into
new systems that do function better and
the the long-term
aim of this r d program isn't
necessarily that
the world will be using these kinds of
systems another
option another you know potential
future is that current governance
systems current
legal systems current financial systems
you know
uh all of our current systems might
learn from these experience

these experiments everywhere and go you
know what that's that's actually a good
idea that works well let's
incorporate that into our structure and
uh you know we'll be better off everyone
will be better off so that would be fun
maybe that maybe the clubs
grow to a point around the world and
they just
they the ideas just assimilate out into
the larger world that are taken up and
then everyone is still doing
better so that would be great or maybe
it doesn't go that way maybe
maybe it becomes a you know like us
against them in a sense there's the
clubs that are growing
the the native systems the current
systems are not doing well
they refuse to learn from these
experience experiments they're they're
not interested in change
well then they probably will be
dinosaurs you know
i mean i'm just thinking about uh so
there are examples of um
a community energy generation for
example so those solutions
start to exist um you know and then
there are examples
of invested interests where they've kind
of seen that as a potential threat
and move to outlaw them so i mean it's
just those kind of you know it's it's
those kind of conflicts
i was wondering where they might yeah
you know um of course those kinds of
conflicts will happen

um and the and the r d program and the
people involved and creating this whole
system have to think these things
through
carefully to try to avoid you know
problems that might be foreseen right
obviously so these are very important
questions and
we would have to give a lot of thought
as to how we
you know skirt past these problems now
you know and and i you know i don't
that's to be done
that's work to be done right i have a
few ideas
and that's and i and i can say that's
one of the reasons i wanted the global
science community as a kind of a
you know co-leader in this whole process
because
it is for one thing it's global right
it's not
limited to the laws of a particular
nation
so and it's authoritative in the sense
that
we can write papers and we can do
simulations and we can
talk with some authority of it would
this be a good
you know do these things help humans
would this be good for human
societies and civilization you know we
can we can do we can do that we can
we're independent of any particular
nation in a
sense um
and then suppose that there's a nation

that says no way this is against the law
you can't do it here well
okay you can't do it there but there
will be other nations that
wouldn't have those rules and this club
model can thrive wherever it's allowed
to thrive right and
you would think that there would be some
nation somewhere in this world where
there might be some allowance for these
kinds of social experiments to occur
and once it works that's the thing
that's the concept is that once it's
shown to work
you know first in one community say and
then maybe in two and then four and then
eight
you know once it's once it looks like
this actually works
then there's a whole nother slew of of
papers and simulations showing you know
like working
you know assessing that and that
information is broadcast to the world
and uh you know at some point it becomes
harder and harder for a society to
to keep it out because
you know no no no country
uses typewriters anymore you know
except the hipster countries so we have
except they have except the except the
hipsters
they'll say i want that good
old-fashioned government so
scott and then a question from chat and
then dave
so um great conversation thank you this
is so

fascinating so the couple of book
references and a couple
of things i'm going to talk a little bit
about money and risk here going back to
what we were talking about a little
while ago and that moved from
money and prosperity you know there's a
book by i think it's hannah's men called
doing money
where the author suggests that money is
a universal risk
consolidator and so i'll hold up here
20 billion okay okay
due to the bearer upon react so it only
is as good as the narrative
right so um i always carry zimbabwe
money around because that's inflation
protected in the united states
we can't get near the inflation rates of
this so um
the the reason i raise that is money you
know we there's a book called the past
as a foreign country
where we think people in the past oh
they just didn't get it and
then we look at this and go you know
money actually
it's interesting when people want to
accumulate money there's a lot of
theories of money this
smith and wixel and hume but money is a
form of communication between people
it's an encoding of risk
in a sense right when we look at and
decoding of risk and a universal
risk encoding in a sense and
it's kind of interesting because then it
becomes abstracted into its own

get its own end that's when it
comes the commodity of money right and
so it's not a medium of communication in
exchange anymore but its own
gain its own because it's used
generically against all risks i can
de-risk
so many universal risks that's pretty
true it's like a universal solvent of
water
you know it becomes something that's
biologically very dynamic
and so the idea of the medium of
exchange is a fascinating one here
so i just wanted to fuss with that a
little bit and then so if you look at it
that way
then the meaning
so this is zimbabwe local community
meaning
and then there's united states local
community meeting and they're different
because the community is what generates
the value not
the money this is a piece of paper has
exactly the same value as a us
dollar but it's the symbol of a system
and the system is the community in the
communities where the value comes from
not from the piece of paper so
that's the intrinsic uncovering one of
the things that i've been
fussing with just to share on this
point is so the past people have been
doing money and maybe they're not so
stupid maybe they've been doing clubs
maybe this is a money club right
um and so one of the challenges is well

now maybe we need a redistribution idea
among the clubs and i think one of the
things that's
really appealing about the way you're
describing things is the intrinsic
transparency
that comes with the exponential increase
in interaction volumes
so it's impossible to su press
transparency among so many interactions
is very similar to after 2008 financial
break
where there were calls to get rid of the
rating agencies because that was one
stop shop for conflicts of interest you
pay them more money you get a better
rating
and they said instead let's get rid of
the
[Music]
trading agencies and let's use the
spreads in credit default swap markets
because it tells you the same
information will the company pay its
debts as they become to that's what the
rating agencies tell you
and that's what the spreads and credit
default swap markets tell you but the
difference is in this in the credit to
falsehood markets
those are set by thousands of
interactions so you can't game them
right so that's what distributed
governance looks like and that's what
intrinsically you're it feels like
you're building that from the bottom up
but not just vis-a-vis financial
instruments but vis-a-vis anything

that community as the source of trust
and knowledge
and then the i call it the community
trust community information boomerang
because then when you want to
authenticate or audit you go back to the
community because that's the ultimate
source of trust because that's where it
came from the first place because it
didn't come from the data passing
through it
it came from the fact of the meaning in
the community
drenching and enriching the information
as it goes through right so i really
love the way you're talking i just want
to make a
few comments on it but that abstraction
of money
is kind of an interesting thing that you
get to levels of abstraction
where abstraction itself becomes a form
of violence
because it's because it reduces the
granularity with which a system
sees the individual and so just last
point
kandinsky of the folks on this phone
have heard this already but kandinsky
made it
the painter said once that violence
societies yield abstract art
and one of the things i always wondered
is that reversal is abstraction itself
violence
and to me it feels like getting to a
certain scale
it feels like violence try dealing with

verizon the other day i
was so frustrated i was wanted to slam
my phone down on the receiver down but i
only had a cell phone i don't have an
old style phone so i couldn't slam it
down
with drama anyway
so these are you've raised a whole
series of really interesting and
important
points so
first what is money
you know if we're going to if we're
going to design new systems including
new economic systems new financial
systems new monetary systems
what is what is money what is what is
what is what is its function
in a in a club or in a society right
well um already in what i call native
systems money is a voting tool if if i
have a billion dollars
i get to vote for any number of things
that i want i can make
i can make whole organizations do
fulfill my you know desires because i
have
i have enormous voting power now
obviously that voting power is not
democratic
it's not transparent it's you know it's
really a dysfunctional
uh inflow of information uh and
the and quite obviously if we're going
to design new systems we would want to
it's fine that money is used as an
information tool
but that information tool should be

completely transparent
fairly distributed uh you know it should
actually function
as a way for all of us to share
our desires our fears our our
our you know our viewpoints our
our preferences to the community you
know it's
redesigned money that serves an
informational purpose and is inter
and is transparent and is you know
deeply
democratic in its function that's a you
that that would be a useful way to
communicate you know
uh so so that's part of the task is to
how do we do that how do we design
a system that functions like how do we
design an economic system that functions
that way
where money is no longer even viewed as
money in the normal
way that economics is as a matter of
fact that economics the like the
economic profession
say for a club is not even resembles
economics because suddenly we're not
talking about economics we're talking
about cognition
we're talking about a societal cog this
is all about
decision making under uncertainty
in reference to some fitness you know
vitality
score of like how do this is cognition
how do we
how do we it's no longer economics
now it's a now it's a new field called

cognition that happens to use a voting
tool called that we might call money
a couple things on that so um you know a
club can't redesign the same national
currency can't redesign the dollar or
anything like that so but it can create
a currency
for its own use within its own within
its own
milieu that functions exactly the way
that i'm describing that that is
transparent that is
you know does all the things that we've
just been describing
so i take a shot at that in my design of
the lead
framework and um if we get to it
i'll try to you know show you how kind
of conceptually how that works well it's
just
just a quick follow-up that's one of the
elements of sovereignty right is the
tax and the money issuance the m1
supply you can't use a starbucks card
everywhere because that would intrude on
the sovereign supply of money on the
fiat
and so the exertion of authority into
every every interaction
is has three parties in it because the
trading medium is a declaration by the
sovereign
that they are there to help but it's
like
um von munchausen by proxy syndrome
right they they keep people weak in
their interactions in order
for themselves to be strong

right very interesting just because
there's a bunch of questions the chat
and a lot of raised hands so i want to
get to everyone in order
so um dave and then i'm going to ask a
question in the chat and then blue and
stephen
yeah it's not self-evident what the
relations
of property intellectual property
occupation competence uh
in the u.s there seems to be a um
an insistence that there that the
the traditional money relationship
structures everything else your
competence
is determined by who writes your page
your credibility is determined by that
um it isn't that way every place
in mexico um the right
to professional association and
voluntary association is written into
the constitution
in the philippines the government is
constitutionally obligated not just to
tolerate
but to actively promote professional
associations and voluntary associations
uh and reading some of the things and by
no means all the things that robert
david steele
says denmark apparently is
doing a fabulous job of managing the
knowledge economy
and uh moving more and more all the time
to
open source everything except maybe
initiation of government force

out and out violence to preserve
buildings and children and things like
that
thanks for the comment sure yeah you
looked at
it steel's uh proposals for
um setting up a very much more
transparent way of exchanging knowledge
and vetting competence
no i'm not i'm not familiar let's um
share that it sounds interesting though
yeah
cool and and part of this prelude in the
research and development phase is
certainly to
take stock of various proposals that
address
different aspects related to
transparency accountability
information propagation value exchange
crypto economics
nfts all these things are going to be
woven together in a new
tapestry that hopefully is our parachute
slash trampoline
so yvonne
yvonne asks a question in the chat he
says hello thanks john for the second
stream
sixth stream it is excellent my question
is
does the club model imply a governance
where two or more clubs come into
confrontation in their ways of
acting so early stage club development
or late stage let's just say two clubs
don't just believe different things they
don't just infer

differently but they act differently
what do we do
sure okay so um so
the in the r d program maybe the first
task will just be to work out how
cognition
societal cognition can happen in a club
setting you know like well
and then all the different systems that
support them
but very quickly you know once progress
is being made towards that end
the next big question is how do clubs
cognate
how how do they communicate
so just as there would be just as there
would be systems and technologies and
frameworks
that allow individuals in a club to
communicate and cognate together and
make decisions together
whether they're economic decisions or
whether they're governance decisions or
whatever kind of
legal decisions whatever that is there
would be a similar
kind of overlay to the to the
between club setting right so that those
those
maybe it would be similar technologies
maybe they would be adapted a little bit
for the club to club
interactions but the same kind of
concept
holds is that uh clubs
the community of clubs is a cognitive
community is a
you know is a is an organism a cognitive

organism
so how does it make decisions and view
the world
and uh and act
now it might be that let's say that down
the road there are
hundreds of clubs or thousands of clubs
and one club
says you know what we
really like this slavery idea that's
gonna work really well for us and you
know we're gonna
we're gonna go off on this tension or
you know some
some say destructive tangent um
well you know i mean how do you how does
a healthy
family or a healthy group of friends
deal with issues like this you know how
do organisms deal with issues like this
you can put pressure in various ways on
an organism that is being anti-social
you know or an individual in your family
a brother that's being anti-violent or
anti-social there's various ways you can
try to pressure them maybe to
behave more socially appropriately or
something
uh so that so that would be part of the
the conceptualizing and the design of
these frameworks is how do you deal with
the
how do you deal with problems when when
some club is not is you know misbehaving
or some individual and a club is
misbehaving there ought to be
that ought to be well thought out so you
have a procedure that you follow when

things like that happen right
but but just off the top of my head
there would be many things you might do
say a club was you know somehow turned
rogue like a cancer
in this community well you can you can
decide not to trade with it you can all
the other clubs can say you know what
this is not
this is not part of our larger community
this isn't right this doesn't work
and we're gonna exile you you know
i mean various things you might do but
all of these have to be given thought of
how do you deal with the situations that
arise
and uh you know there there could well
be
reasonable ways that the communication
through you know through other means
that you could
start to harmonize the whole if it's
in a state of disharmony thanks i mean
obviously
obviously the human body does that right
i mean the human
there's trillions of cells in the human
body and somehow
they basically all work together and on
a rare occasion
some cells decide not to the the
and we might call that a cancer so you
know that cancers can happen
and how do you then put you know the
question becomes how do you prevent
cancer from happening
and obviously what i would say obviously
one way is

keep the organism healthy you know keep
it keep it all
healthy and
that helps um keep communications open
you know that that helps and then if
cancer still arises then you know how do
you then we have to have a way to deal
with it
thanks john so blue and then stephen
so john i just want to say thanks i've
really enjoyed all of your streams
thus far and this is like a huge problem
that um you've attempted to tackle here
and so i don't want to like
um approach it in in a way like it seems
like i'm a naysayer or like i'm
boohooing i just
there's some extra components here that
i think
um you know maybe you have thought about
or have approached it in a very delicate
way so the concentration of wealth and
power power i i feel
in this country and this is definitely
personal opinion or maybe in the world
it is it like stems from some kind of
spiritual bankruptcy
right and and we've always had authority
and like you talked about authority and
societal cognition
in the last paper and you talked about
you know how authority comes from either
voluntary deference
or coercion right and previously
from you know the beginning of written
record until now
and maybe even still now that voluntary
indeference and coercion both can come

from
religion and religious groups and we've
seen the god gene
right and so i know that like you've
kind of sidestepped and produced a very
agnostic
platform and and i mean i know that you
also did talk about like the
interconnectedness
of all things and kind of um bringing
that forward in
an education way and not a religious way
and i
i definitely appreciate that but do you
think that they're
it's possible that we're hardwired you
know we have this god gene right like
that we're hardwired to need
something beyond and that that if we
have that something beyond
maybe the concentration of wealth and
power if we all gave
you know if we all did tithing 10 of all
of our salary
i mean i think that that would equal out
the world just that
right so so i mean is it possible that
religion is
absent and maybe plays the more central
role in
society than you've allowed for here
yeah i've not talked about religion much
at all in
any of the papers nor in the series
these discussions and i don't i don't
really want to
um i i think this
program can go and continue it can move

forward
and reach its aims fine without
without focusing on the topic of
religion
now
you know because people people belong to
all kinds of organizations including
religious organizations they
people get meaning from all kinds of
organizations
and associations so good you know fine
that's
fine the the topic here
is how does a community
cognate how does it make decisions how
can we create
systems that favor beneficial and
healthy cognition
right so those two things can exist
at the same time you know people can
belong to religions and
um and and participate in healthier
systems
at the same time um as far as meaning
goes uh
you know people find some people find
meaning in religion
um but in the uh
i think it's useful for this project to
also
um consider the the some of the ideas
that i had in my first paper world view
because those ideas of like how we are
all interconnected how we
are like trans really in in some sense
transcendent ideas that that have
scientific
you know a rationale

behind them right um
that's useful to consider those points
in the worldview
paper because that at least in my
in my thinking those concepts are
are central to even coming up with
the the uh concept of purpose
so what is the if we're if we're
together in a society
what are we doing what is our purpose
like if we don't have a purpose why are
we even
associating with each other you know
just randomly or like
just to take advantage of each other
like why are we even in a
community together you know like whether
it's a large community or a small
community
so i'm suggesting in the particularly in
the first paper that
that if we come together with a purpose
you know and that purpose is to thrive
now and in the future
then we can do something about that we
can you know we can create
systems that allow us to fulfill that
purpose
that are true that are true to purpose
the technology in a sense that's
that functions the way it's supposed to
function um
and the the concept of purpose comes
from this larger world view of who
what is a human what is an individual
how
how are we all related to all of life um
you know that's

at least my mind that's kind of where
where the idea of purpose comes from
like you know that we're
actually together as a society we're an
organism and an organism is connected to
all
other organisms and all of life is
connected all the way up to the
biosphere
and our purpose and include all of that
so
you could call that a trans you know a
transcendent
perspective perhaps and that might be
i'm i'm hoping that might be meaningful
to
folks who might want to be involved in
this kind of project
because uh you know if it is there's
scientific reasons
for all of that you know there's a
rationale for all of that that
we can explore all of that
scientifically and otherwise you know
uh so so yes people
belong to religions and they can
continue to belong to religions and
and and from my perspective that
that you know those organizations will
will
continue to exist as long as they're
useful just like any organization just
like any association it should continue
to
exist as long as it's useful it's not
going to happen in the club setting
you know some people might be
religiously oriented that's fine

and but that's not really the you know
the the
the club focus the r d project the
the design of new systems the engagement
of people in a societal cognition
framework that can all happen whether
someone is religious
or not i would say so so just as i said
did i sidestep it well enough there
well can i just ask a quick follow-up so
so
so definitely in the um societal
cognition i mean
i just think that um we have to factor
in
the role that religion plays in the
societal cognition because
we give our we do voluntary deference
and coercion through religion and and i
think that it's something that if people
ascribe to a religious practice that we
have to
allow for that and examine what role
that will have on societal cognition at
large like whether they're part of a
religious organization or not
or whatever oh sure those those are fine
questions does
does say you know does religion a in
some ways
impact the capacity of these systems to
function in the way we're thinking they
could function that's a fair question
that's a scientific question
or perhaps examining the the framework
through the lens of each religion would
be maybe useful
and that would be useful also how that

would be part of the question
of like you know is there some impact
that this that religion a is having on
this system
and uh if so let's talk about what how
do how does this group
see the world how does that how does
that you know how do
what's our differences of the way we see
the world you know those are all
fine questions and all of those
questions can be addressed in a kind of
a scientific manner
when um when scott mentioned that
there's the buyer in the cellar but also
there's like the umbrella
of the trust in the sovereign in the
fiat you know
fiat lux uh is how the book begins in
some ways
it's a lot like a marriage partnership
where there is a third party and
that third party is the context and
various religions have different ways to
talk about that third entity
but it's so interesting how when we're
dealing with identity and communication
and
trust it is like we need to have a
higher level scaffold
now can nature provide that can
so-called science today or in the future
provide that
that's what we're going to discover and
discuss
so um stephen and then lee and then dave
yeah this this idea of
the environment and how we connect in to

say a shared epistemic niche
a way of understanding and
kristen talks about that in some of his
interviews
around you know even religion can help
us have a shared
i think it's an epistemic framework or
an epistemic niche for
how we can align and that
then ties into you know how do we
treat hidden states if we're going to
have active inference so as well as
cognition
dynamical system in the brain and body
you know the niche
and where we are in the niche becomes a
big part of that
and of course what we've seen over the
last 200 years is
there's more and more states that are
being fixed
in the environment the things that we
say we know or mechanical
objects which will take over and get rid
of
uncertainty markets will supposedly deal
with all of that
all these different um and it leaves
less of a role for trying to attune to
hidden states
beyond so that that may be
um one of those questions around
you know money being a classic case of
that you know how much do we trust
in the econom economics and that will
solve
you know just vote for the best
economics and it will deal with

everything
um but where is that is that really a
deontic queue like fristen talks about
deontic queues are accusing the
environment like traffic lights
that we just build all the traffic
lights and then we just work
to do what they tell us but that's where
that violence can come in that i think
was also mentioned is because
deontic cues will always win if they're
very strong
over active influence in a community
because active influence in the
community needs integration it needs
time to actually listen to the landscape
and if you can have deontic cues that
say well
just you know that's not needed you kill
a lot of that so in some way
ways that might be relevant here is
showing the benefit of epistemic niches
yeah
in some ways these conversations are
related to
some of the points that blue is picking
up on
so so you know um
if you try to to talk a little bit about
about this so religion is a part of
culture
you know you can think of that as
religious groups as a different cultural
groups kind of social cultural groups
right
um you know culture is really useful
uh no cultural norms are really useful
can be useful they can

draw our attention the things that in
the world that are most important and
help us to avoid danger
and you know healthy culture is
is a part of cognition it's a way to
offload
my cognitive burden onto the you know
the the
history of this culture that has learned
as it has gone through life what works
and what doesn't work you know
so um so
[Music]
so all you know it is good that
that say we had a large club with
multiple
multiple subsets of cultural subsets in
it
the the the again the purpose of the
club
is societal cognition right and we we
should have ways to measure
if that's functional or dysfunctional
both in
are we achieving a quality of high
quality of life are we
you know are we are we is our
uncertainty about the future high
or low or you know how are we doing
so so part of that is is engaging
with cultural groups of subsets you know
within a within a
club or within a community right the
idea of
all of this what you know like what is
the idea of cognition well
cognition requires communication
there's no that cells in the brain don't

function
if there's not communication that's
going back and forth rich
rich communication right so we can
decide
as a as a as a group you know we can we
can
we can we can study different kind of
designs for systems we can measure
their success we can you know we can we
can apply
a kind of a scientific method to come up
with good ways to ensure that we're
we can all thrive but part of that is
also uh allowing for
and providing for and facilitating the
communication between
this group associations that are subsets
of any club or any society right i mean
the
idea has to be communication and in in a
cooperative
sense right or else there's no cognition
to begin with you know then you just
have rules
and it's who god knows who made those
rules in the first place you know
so we're just all we're doing is like
just just look at how
cognition works and let's be honest
about it let's be transparent about it
let's you know let's just peel it peel
back the
covers and see how social cognition
works and see how cognition works
in biology and learn what we can learn
from it and and
uh let's bring the richness of that out

and make it all transparent and fair
so that allows for subgroups to have
cultural differences that a lot
that's important because cultural you
know
cultural subgroups will have a different
perspective on
how to do things or what's important or
where we should focus our attention you
know that's
all good right as long as as long as
it's set up such that we're all
able to openly communicate
and tell our story and tell our
narrative whether that's as an
individual or whether that's as a
cultural subgroup you know there's a
narrative to express to
there's concerns there's desires there's
needs you know again humans are very
complex organisms we have very complex
needs
and all of those you know all the core
needs need to be fulfilled in whatever
healthy society we build
the need for association the need for
love the need for
helping each other like we you know we
want to help each other
we have to address all of that and it
just has to be open and transparent and
openly discussed about what we're doing
and how this works and how people
you know how how these tools and
approaches facilitate
cooperation and cognition or hinder it
you know
that's the experiment thanks john

lee and then anyone else with a question
uh yeah i think this is a a related
question um i'm just going to read out
just one thing from your paper actually
because um
so you in this paragraph you uh
you say um
in a second-order approach to active
imprint um
if a society understands itself as a
cognitive organism
uh if it understands its cognitive
process as reducing uncertainty for the
purpose of achieving and sustainably
maintaining
vitality and understand its social
systems as a cognitive architecture
then given those beliefs what kind of
societal systems might
it develop and employ um so that got me
thinking
is that are those beliefs
prerequisite then for everybody in the
club does it need to be that level of
shared understanding or can it be just
held within a
kind of a subsystem you know is it is
that something that perhaps the
the scientists who are doing the
simulation etc right right
that's a good question now um you i
think we should uh
allow that if this r d program were to
go forward that you know it might start
small and focused and there might be a
focused group involved of those
people who are interested in this kind
of thing right and if it started to

work and you know it proved to be
beneficial to communities then it might
grow and other people would get involved
and
so we can think of this whole thing as
progression i think that's fair that's
the way it would
naturally play out and uh
you know if the scientific community is
involved in this and if the scientific
community is involved in creating
and helping to you know co-designing new
systems well then there's going to
certainly be
a focus on the scientific viewpoint
in order to do that communities
obviously would have their input into
the r d
program and you know it would between
the two between the two
it would move forward but i would guess
though
that early participants and early
efforts
would be uh focused you know would more
or less align with what i have in the
worldview paper
or else they why would they be
interested in participating
in the first place you know like they
would maybe only be interested because
they'll they can see some alignment with
the goals of the program and
the way the program is oriented and
what it's trying to achieve right so
quite likely the
early participants in this whole thing
whether they're scientists or community

members might be kind of more or less
aligned with
ideas in the world of your paper and and
especially the purpose
of you know the purpose of a club or the
purpose of societal cognition
now suppose down the road you know 20
years from now i live in los angeles and
they're you know like i don't i don't
care at all about
any of that stuff i just want to live my
life
and you know make sure my i have a roof
over my head and stuff
but this club is up and going and is
doing well and they're all there
you know they're actually doing well my
neighbor is
a part of it and his life is improved in
some way
so like i don't really i don't really
you know like i don't i don't care about
any of the
purpose stuff too much but i like what's
happening you know like they got a
nice scene going on there that's
beneficial so i'm gonna join anyway you
know
and it's so to a degree the the culture
of it would change it would evolve
but hopefully that would be it would
start with a certain
kind of it would start in a healthy
functional way you know aligned with the
purpose of the whole project
and then it would evolve the way it
evolves you know i would
i would guess that if it's beneficial

those beneficial ideas would continue to
be the
you know at its core somehow but how it
exactly it evolves you know as as
decades goes on and as you know as
is 100 years from now it's maybe maybe
you know we if this exists 100 years
from now and if it's spread widely maybe
we're
using very different language to talk
about it and
we see it differently perhaps
thanks john um now as the author
it is up to you would you like to shift
the second slide
for today these are awesome
questions thanks everyone in the chat
and for
participating because it's it's a
conversation and john's
uh initial inroads are tripartite
and very provocative and we're seeing
that play out and so it's a great
beginning to a discussion even though
it's our dot 2 video so john where would
you like to take it for the last second
uh yeah obviously there's a lot of ideas
in this whole series there's just a ton
of ideas
uh both both uh superficial and deep
and if this goes forward it's gonna take
a
lot of discussion and it's gonna take a
lot of people involved to bring
different perspectives to
kind of make bring this whole concept to
fruition right so
we're just scratching the surface that's

all we're doing we're just
putting out some of the big ideas and
then you know there's a ton of work that
has to happen to make this real
so uh can you go to uh four
number four group four slides maybe uh
um maybe uh slide
seven six if uh seven integrated
components on slide six
six yeah yeah seven integrated
components yeah awesome okay so
uh maybe in the last little remaining
time we'll just touch on this
lita framework the simulation model that
i put out
it's really the the idea of the leader
is
more than just an economic system it is
actually
the idea of it that is put out in the
in uh in my book economic direct
democracy
um talks about the whole picture the
larger picture but there
there's as you can see from the slide
there's seven components to that
and and they really touch on all six
overarching systems
that i talk about in the series so
economic governance legal
analytical education and health it's all
in there somewhere
so the idea of the leader that's local
economic direct democracy association
the idea of the leader is an integrated
framework of cognitive systems
the simulation really focuses more just
on the economic aspect because that was

the eat really was just the easiest
piece to pick up first um
and uh the the the uh in particular it
focuses on the
what i call the token exchange system
which is the first three in the list
so there's a token is a is a local
digital currency
so um that the community creates and
uses
uh there's the token monetary system
there's what i call the crowd-based
financial system
uh and there's a market system local
markets people sell you know creating
goods and sell them in local markets
they also obviously sell
distantly to people who are uh
either in the local area but not a part
of the club but also
far distant that are not a part of the
club
so those uh first three things
constitute the token exchange system
and we'll just maybe talk a little bit
about the simulation and some of the
results and
how that works um
maybe slide eight yeah yep
so in the simulation there's five
aggregate agents uh there's
organizations
persons the cloud-based financial system
local government and then the rest of
the world which i call
rest of counties the the data for the
simulation
the initial data was from the u.s census

for eugene oregon
relatively small community a small
county population 100 000 or something
like that
and i used starting income values from
the census data for
eugene county from 2014 or something
like that
um this paper was published i believe in
2014 so i probably used a little earlier
data
to get things rolling okay so
the the simulation just it really
illustrates
it doesn't it doesn't predict how the
system would work it just
illustrates my thinking about how a
system could work
and in particular i wanted to uh i
wanted to
design a system that would that could
bootstrap itself up
without large influx of cash
or other things from outside so that was
part of the idea how can a system
bootstrap itself up
start with very little and drove from
there the second was i wanted to create
an economic system for the
for the club model where over time
incomes became
equal income income and wealth equalized
and
the the thinking there is that
money and and and you know the whole
economic and financial aspects of that
really is part of the cognitive
community's cognitive system so we're

using money as a very transparent voting
tool right
it's like votes information it's
carrying information about people's
desires and
fears and everything um
uh yeah so that's uh
that's i wanted to show that using a
stock flow consistent model agent-based
stock stock flow consistent model
i just wanted to show that the numbers
add up correctly
right like i'm not i'm not i'm not
creating uh
dollars out of nowhere there's like a
there's a logical way
this plays out and all the numbers add
up there
if there's a flow from one agent
uh one of these major agents there's a
you know there's a
any flow from one agent goes somewhere
it just doesn't disappear
uh so uh very briefly the
uh the lines in blue
are the flow of tokens and dollars
so um a community is creating tokens as
it goes as it grows it creates
more tokens the the concept is that the
value of a token would be sort of
it's sort of equivalent to the value of
a dollar
and it's made to be non-inflation
inflationary so
let's just just say for 2014 it was the
value of a token is the value of a
dollar
purchasing power of a token is that of a

dollar in 2014.
and uh um the community would create
tokens
a little bit in the first year a little
bit more of the second years more people
get engaged there's more use for tokens
there's more ways to and you know
use them in in the local markets then
there would be more tokens
tokens created and that and the and the
volume of tokens in the local community
would grow
and salaries for those who were members
of the club salaries would be paid both
in tokens and in dollars so maybe in the
first year
you would just get a few tokens and then
over time your share of tokens
in your wages would go up and the same
would be true
even if you're just saying you you
yourself own a business you know people
you would sell your if you're a member
of the community you would sell
things to other members of the community
and they would pay you
in both dollars and tokens amusing
dollar as the national currency
so organizations just to look at the
blue lines again
organizations pay wages to persons the
persons might be employed or
unemployed i i also call unemployed not
that there's
some people who are unemployed or not in
the workforce that's what an if niwh
stands for
and uh people who members who receive

dollars and tokens as part of their
wages as part of their income
would naturally they would
have to contribute a certain fraction to
the crowd-based financial system
and uh that's the blue line for
contributions to the cbfs
and the cn from the cbfs if i'm an
organization or i'm an individual and i
have some idea to start a
business or i need i need loans for my
business or
i'm a non-profit and i want grants i can
go to the cbfs
and apply for a grant or or a subsidy or
donation
and once as a person
once my once some of my currency is in
the cdfs
i more or less get to decide how that
money is used like i want to support
schools or i i like this idea of a new
light bulb i like that
i like that business idea that this
person came up with i'd like to
give them a subsidy or i'd like to give
them a
loan or a donation or something like
that again this is all very transparent
and and again as time goes on incomes uh
become more equal
so that and then because incomes become
more equal
everyone has equal amount of of power
in the cbfs to direct how the community
spends its money
the green lines in this figure are
dollars

and obviously you have to pay taxes and
dollars the
the government's not going to accept
tokens so um
uh you paid all taxes and dollars the
government also provides some income
support for people who are unemployed
that's
natural normal the government also
provides some grants and contracts and
subsidies to
organizations that's quite normal and
the
organ organizations that our members
also have to deal with
the outside world they have to sell
their goods and and
exchange uh goods and services
using dollars so those are the green
lines
uh so that's how the basic system works
and then i and then as this
this the system evolves over a period of
28 years in the simulation
and i'll show you what the results are
as the as
as everything goes round and round we're
not going to have
we only have a few minutes left so i'm
going to kind of have to get just get to
the point here
there's some assumptions uh on the next
slide
what john can i just ask um which of
these agents are
active inference agents or where does
active inference come into play
or could come into play with later

versions of this simulation
um okay so again the the the um
the leader framework is an integrated
system that has a governance component
if
the governance component has a
legislative component
and you know all the major six
overarching systems are addressed in
some way
i'm only simulating the economic part
right so
so if you were in a real club and a real
club looked anything like the leader
there would be a governance system that
you would
be you know participating in some kind
of collaborative governance system
to help decide you know what should
happen in the community that'd be the
legislative system legal system and so
on
uh education system and everything so
that's one way that active imprints
would be involved just in the process of
decision making under uncertainty right
and in the in this particular simulation
you know obviously businesses are having
to make decisions and choose
choose what they do with their resources
so that's another way that
cognition is happening and then maybe
the main
in this economic picture maybe the main
agent is the cbfs that's a that's a
financial system where everyone
is able to participate in a transparent
and democratic way

to fund those things that they think are
useful to their community that create
the kind of jobs
that they want to create to you know do
as a community do we want
lots of non-profit jobs do we want
for-profit jobs do we want
jobs in the manufacturing center do we
manufacturing sector do we want you know
like do we want to support this kind of
business do we want to support that kind
of business
these are all decisions that the
community makes through the cdfs
and then you know if if the community
decides like yeah we want this bakery
this would be great
or this bookstore or this farm or
whatever it is
then the community can you know
associate within the cbfs and say okay
there's a thousand of us we like this
idea for this farm
here's funding for it so you know go for
it
so active inference particularly would
be happening in the cpfs
yeah steven a question and then anyone
else with a question
yeah i suppose in a way you've seen a
mix here between
like in the in the red boxes you could
have active inference
like the dynamical non-linear systems of
inference and then you've got this
message passing
which to some extent is then the
aggregated

system the sort of steady equilibrium
system
i the pile of money or the spreadsheet
and a message passing where data goes
from one
to another but then within that
there may have to be another process of
inference that happens assuming people
have any agency and it's not just
all automated so it's like to me that's
probably where you've got
system dynamics in that diagram and then
you've got these
inside that non-linear dynamical systems
i.e the inference processes embedded
inside
yeah there obviously is a lot going on
there that we're not going to have time
to talk about because we have like five
minutes left or ten minutes left
but but just to give you one example of
that
suppose that a a a group
in the community wants to create some
kind of business and they want to
produce some kind of product right
so they come to the cpfs and say hey we
want to make this new thing well
a lot of information has to go back and
forth right this is an information
rich system so i wanna if i'm gonna
if i'm gonna support this with my
resources in the cbfs i wanna know
where you get what you know where are
you getting the raw materials
how is that impacting the environment
how is your product going to improve or
harm my our sense of community or

any one of the metrics that we're
measuring
how is that gonna is there any pollution
how are you gonna deal with your waste
how much energy do you need
uh do we have that that level of energy
like electricity or whatever
as a community do we have that resource
about i mean there's an enormous amount
of discussion and information has to
flow between these
different parties in order to make the
kind of decisions that you know
intelligent decisions that you would
want to make in the cbfs
okay so um in the few minutes that we
have
remaining if you could turn to slide 15
please
all right go for it so these are some of
the results um
uh um
and i should say that um some of these
that right at the get-go as the
community is forming as a club is
forming you know
they would be choosing their trajectory
they would be and and this upper
left-hand corner is an example of that
they would be choosing that they want
income to rise to a certain level over
time that
they want participation to happen in
this club at a certain rate over time
they want to uh they want in particular
the lower
lower left there they want the income
target to rise

at a certain rate over time and um
and obviously you can't ju you you know
like you can't just
flood a community with money on day one
and expect that to work
because there's no the same with uh say
with digital currency
because there's no no place to spend it
like there's you know it's a tool for
voting and if there's no place to vote
the tool is useless right so that's
partly why things grow slowly over time
so that there becomes
outlets for using the currency it you
know and
to create the kind of economy and and
economic system that the community wants
so
some of these things would be decided
ahead of time like here's what we want
as a trajectory here's what we're
here's what we're doing as a community
using the system
this is where we're going this is what
we're going to do this is how it works
and if we all
cooperate in this way that's exactly how
it's going to
happen you know that's that's the
concept is so it's not just
here's a system use it however you want
let's see how it goes
it's really more you know we we're a
community that we have a goal
and here's how we intend to reach our
goal we want
say for example we want incomes to rise
to a certain level over time

and this is you know can this system do
that
and if the answer is yes well then
here's how we're going to do it we're
going to
increase it a certain amount every year
and it's all going to work out
and and so in that sense some of this
should be no there should be no
surprises for the community as they do a
field trial it should be
working exactly the way they intended it
to work bringing them where they want to
go for example when you drive a car on a
trip you you know where you want to go
if you're halfway there
if it's a two-day trip and on day one
you're halfway there then you know
you're moving in the right direction
it's all going just where you intended
to go
okay so um on this slide
i may be on the lower right hand corner
uh this is what happens over 30 years as
the simulation moves forward
uh the this is showing average family
income
in the in the membership it starts at a
fairly low level this is from the census
data this is where they start
on your zero at your zero and every year
the average family in income increases
essentially the leader is paying people
to join
either because the because the income
target is rising
and my income is go my wages are going
up as the income target goes up

or i actually have make a lot of money
but i'm joining anyway just to get that
bonus
uh you know whatever it is a thousand
dollars a year in tokens or whatever
that bonus is
so people are joining because it is it
is
in their economic best interest to join
and as
it goes on the income target rises
and then family income you can see the
blue line there
family income average family income more
than doubles over the
30-year period so uh the average family
is
between tokens and dollars the average
family income is over it's
around 110 000 annually as as time goes
on
now i might note that the dollar
aspect obviously the community needs to
bring in dollars in order for this to
happen because some of the
income is paid and some of the wages are
paid in dollars and some are paid in
tokens
so part of what this community is doing
is they're adjusting
their outflows they're adjusting their
transactions with the rest of the world
such that they're keeping dollars in
circulation
longer than would other but otherwise be
natural
so for example there's a like a buy
local program

and there's ways to support uh local
local businesses and keep money keep
dollars circulating in their local
economy longer
and in the end what happens is at this
high
income level what what actually happens
is
the community is is is increasing
the the number of dollars in circulation
locally
up to the point where their
average dollar income is the national
dollar
in average national dollar income so you
know
like today in today's world the incomes
are very very
uh unequal there's
a few extremely wealthy people and most
everyone else is
not making very much but there is an
average dollar
you know average dollar income over the
country and what the leader is doing
over time
is is kind of pulling dollars into their
local system such that
they're just reaching the average
overtime
and then they so they're reaching the
average
average dollar income which is about 70
000 or so dollars
and then they're also have on top of
that uh
income in tokens which the community is
creating so between the tokens and

dollars eventually they get to the point
where
where they're having about a family
income of about a hundred and ten
thousand
thousand uh so
somebody had said earlier that you know
maybe it was you daniel that i'm both
i'm both saying that money is a
different animal in this system and
motivations are different and in the
next breath i'm saying
oh this is a motivation you could you
can make more money
so like both of those things are true at
the same time
and um and there's a transition period
where we go
from here to there right i mean this
whole thing just doesn't happen
overnight if we want to transform
society it takes time and
uh so in this transition period we might
be
we might be thinking of money and income
in both ways like
in the more traditional sense of i need
money to pay my rent and then in the new
sense of
money is a voting tool and uh we all
need to have
more or less equal income so that we
have more less equal voting
power in this community now what happens
happens after 30 years or 50 years you
know maybe our understanding of all of
this deepens and widens and
um you know maybe the the current

concepts of money and economy fall away
over time
in exchange for this new new uh focus on
cognition and well-being and
uh value of community depending on how
well
it's thriving is is this is the code
is the code or the formalism available
because people might tweak a parameter
and get a totally different response
they might include another variable or
another function get a totally different
outcome
so how can we actually version this in
an open source way
so that we can be working together on
something that's transparent
yeah yeah the code was made of open
source but made available
publicly back in 2014 um
and i also there's a
interactive simulation on the website it
needs a little
artwork but it functions so you can go
to the website
principalsocietiesproject.org and you
can
to some degree you can type in some
different numbers and see how that
all works out so you can play with it a
little bit on the website
uh but beyond that um this simulation is
really quite simple
and if i were to recode it i might
record it in a different way today you
know that i did back in 2014
but this the simulation is really quite
simple it needs

this is just a first glance of a
conceptual glance of how this could work
and it needs all kinds of work there's
all kinds of ways that we could expand
this
that you know there's a lot of questions
that need to be answered and we could
expand the simulation to
answer questions better or address more
questions so
uh so you know you can download the code
that i put up
in 2014 but you know ideally i think we
would move forward and
develop you know kind of a more
functional framework for this that we
could try out different approaches
and different ways to do it
um uh maybe on the slide 16
in these last few minutes this is maybe
the one of the kicker slides you can see
on the right the histogram of this is
this
starting distribution of income in the
county
and then below is the
income distribution at the end of the
30-year period and you can see
practically everybody is making this 110
000 in currency per year
on the left you can see how the
unemployment rate changes it starts at
about seven percent
or so uh in the county and drops down
to essentially full employment
on the next slide slide 17
on the top left is the curve of how much
money is in circulation in the cbfs over

time
and you can see that about 6 billion in
currency circulates annually at the end
of 30 years in the cbfs
so again this is a highly transparent
uh deeply democratic and collaborative
way to
decide how money should be spent in the
community
and imagine and this is a very very
small community over like 100 000 people
or something like that so
this is six billion dollars for a
hundred thousand people to decide how
they want to
you know what they want for their
community what kind of jobs they want
what kind of
what kind of uh you know how many
non-profits they want versus for-profits
how what they want like you know that's
a lot of money
and with that much money a community
would have
enormous flexibility and capacity
to make the kinds of decisions that it
would want to make about how how it
lives its life
maybe that's the that's the you know
that's maybe the main
slides the main results essentially
income more than doubles
enormous amount of money circulates
annually in the cbfs
and the community gets decide gets to
decide in a very transparent and deeply
democratic way
how that money is spent so maybe we'll

end there since
we're out of time but there's obviously
a lot more details in the simulation
paper
and anyone can read that to pick up more
thanks so much john so we'll chill for a
few more minutes people can drop off
whenever they want to but maybe if
anyone who wants to raise their hand
uh and give a final thought but it's
awesome that the
code is posted maybe you can add that in
a youtube comment
because it'd be awesome to see maybe
where this even plays nicely with the
active inference models that we've been
exploring
in these recent discussions
so any um one wants to raise their hand
give a few more
seconds for that also one note
about um communication and stephen was
talking about like the
agents active inference agents and
message passing between agents also we
know that
message passing is a mechanism by which
active inference can be computed within
an agent there's a lot of cool papers
on that and it reminded me of um some
work that i had done in the eu social
insects in the ants
which investigated how physiological
mechanisms that were inside
of a solitary insect's body like related
to the regulation of foraging
over evolutionary time in the colony
context

become rewired to be playing out across
multiple bodies
so the regulation of foraging for a
solitary insect is all on board the fly
but the regulation of foraging for the
ant colony is something that's
distributed
and it got there through evolution and
hundreds of millions of years
of it working so yes there are
failures just because one is seeking a
goal doesn't mean that they reach the
goal
most ant species of course are extinct
but it does point to this way
where the kinds of connections that are
inside of an agent
can be of one kind and then the
communications between agents can be
different
we see that with neurons with ants with
societies and
by having a scale free framework like
active inference
we could rethink how communication
happens inside and outside so that we
can take some agency ourselves
in designing our niche in our
evolutionary transitions upcoming
so dave and then lee
um i see some um
data from the actual physical economy
in eugene is there um any interest
in setting up um leon tf matrices
and running an actual physical
simulation of the economy so
you don't just swap everything into into
money but actually use

physical activity and see whether you're
exceeding break even over a period of
decades
i'm not if you mean by physical you mean
like an actual
club yes that's where all of this is
going you know like in in
in whatever would take six years ten
years whenever this is ready
yes let's do a real let's do a field
trial in a real club and see how this
works
but if you're talking about more like
uh in silico sure all i'm looking at
here is just
some simple metrics about about this
change exchange of currency but there's
all kinds of things that have to be
exchanged and considered there's
energy there's wastes there's you know
pollutants there's uh
there's raw materials there's uh human
resources there's
like all of these things all of the
components could become
part of a simulation model and should
become part of simulation models over
time
so again this is a very very simple
simulation and there's tremendous
opportunities to expand it and make it
more meaningful
thanks lee
i am sorry um i'm gonna try and get two
questions for the price of one
so my first question is um given that
there's you know you see this kind of
prelude phase um what

what would you see as uh priorities
initial priorities for
for research then to get this really
moving yeah
i mean good question and and and i wish
i had a uh
you know a clear path to let's do a b c
and d and then it's all going to work
out but i don't so we have to decide
that together really i mean
i could i i could come up with a list of
ideas that we could all discuss together
you know for just one is the these kinds
of simulation models could be expanded
but there's other things too for example
it would be really interesting
to just do a survey either of the
general population in
you know in one country or maybe
multiple countries
or do a simulation of the global in the
global science community and just say
to scientists you know if this
kind of r d program were were available
or were you know it was activated and
there was funding available
would you be interested in participating
in this you know like
it would be really interesting to know
is there support for this
in the scientific community and in the
public does the public
is the public interested in
fundamentally new systems
you know those would be interesting
questions there could be media
some you know somebody should make a
documentary about the concept of

society as a cognitive organism what we
could become
you know like they could be art projects
there could be media projects it could
be literature projects that could be
simulations there could be surveys there
could be
assessments of say you know choose some
community somewhere maybe a smallish
community
and assess what is their uh how do we
apply
metrics to their current fitness what
kind of metrics they could be work on
metrics so they could be work on
applying metrics
to a to an existing community just as a
just to get a background of like where
are we now
and and how would we collect this
information there could be that would
you know there could be work on computer
models to be like
you know if we had 50 metrics how do we
combine them into some
something meaningful for uh you know for
making decisions you know like so
or on the topic of tools how
what kind of tool would allow 10 000
people to richly communicate
and make decisions together and and
how can i give how can i tell my story
to
a thousand people or ten thousand people
how can i can tell how can i tell my
story my narrative my understanding of
things
in some rich way such that it is

computed and used by the whole
in a in a meaningful way there's also
you know a few ideas
you know the right yeah thanks for the
answer and live stream
is a new affordance it allows those to
speak but with a hundred thousand people
there isn't
enough hours in the life to linearly
hear from everyone
and so when we want to speak and we want
to hear and listen active listening
passive speaking all these things how
are we going to make that happen at
scale so
awesome questions you're raising steven
and anyone else and then we'll
close out
yeah i think the scale question keeps
coming up right that's the
start to realize that's that's
everywhere but often is
is not talked about um one thing
just uh sort of to finish off was
there's the inferences
about how to act in the world
which but there's also the ways to get
an insight
into the influences that we're making
so in some ways i always think this is
giving a way to
look more deeply um and go beyond
you know system dynamics which is used
quite a lot in
population work but to go into this
kind of um way to sort of say
okay we're not necessarily creating
these clubs

to tell you what to do and to replace
other government structures but to give
other ways to get insight
into what is going on
in the inference process here yeah
excellent excellent so this is um
you know potential this whole concept is
potentially useful
for making actually making decisions
decisions for the future
as a by a group but also for individuals
and a group to understand
how are we how do we cognitive how are
we cogniting
how can we do how could we cognate
better or how do we cognate how does
how do we actually function you know
like all of these are questions and all
of these are open to assessment
and you know like i could imagine
thousands if this went forward there
could be thousands of papers
on all kinds of topics not just about
cognition but
you know that would include cognition
about how does it actually work how do
humans how do humans associate and
cognite together
what a time it will be so thank you john
it was great to have you on and the six
appearances were just the tip of the
iceberg so we look forward to continuing
the discussion with you with seeing you
in our tools organizational
unit where we're really inspired to be
working towards developing some of those
things that you're suggesting that
there's a need for so thanks everyone

for joining live and for watching and
for the awesome questions yvonne natalia
etc so we'll see you in the coming weeks
with future papers and future
discussions so see you later
thanks much bye